

Optimal Central Bank Collateral Policy for the Net Zero Transition*

MATTHIAS KALDORF[†]

February 20, 2025

Abstract

This paper proposes a quantitative DSGE model with financial and environmental frictions to assess how the transition to net zero emissions affects central bank collateral policy, which specifies the assets that banks can pledge to obtain central bank funding. Climate policy affects macroeconomic outcomes that are relevant for the (optimal) conduct of central bank collateral policy. In the short run, high emission taxes increase default rates in the non-financial sector. In the longer run, they reduce the return on investment, loan demand by non-financial firms and, hence, the amount of collateral available to banks. Welfare-maximizing collateral policy is temporarily tight in response to positive emission tax shocks in order to reduce the central bank's exposure to risky collateral. By contrast, it is structurally more lenient in the longer run in order to counteract negative effects of emission taxes on collateral availability.

Keywords: Central Bank Collateral Policy, Corporate Default Risk, Net Zero Transition, Transition Risk, Collateral Premia

JEL Classification: E44, E58, E63, Q58

*Falko Fecht, Philipp König, Vivien Lewis, Emmanuel Moench and participants of the Bundesbank Research Seminar provided useful comments and suggestions. The views expressed here are our own and do not necessarily reflect those of the Deutsche Bundesbank or the Eurosystem.

[†]Deutsche Bundesbank, Research Centre. Wilhelm-Epstein-Str. 14, 60431 Frankfurt am Main, Germany.
Email: matthias.kaldorf@bundesbank.de.

1 Introduction

How does climate policy affect central bank collateral frameworks and their optimal design? Collateral frameworks specify the assets banks can pledge to obtain central bank funding, such as government and corporate bonds, and the valuation haircut applied to each asset. In practice, collateral frameworks are set wide enough to ensure that the *quantity* of collateral allows banks to borrow from the central bank when necessary, but narrow enough to ensure that the central bank is not exposed to low *quality* collateral if a bank defaults on its central bank loans. This paper assesses the effect of climate policy on the basic collateral policy trade-off through the lenses of a macroeconomic model with financial and environmental frictions.

Our framework allows us to study the short and long run implications of climate policy emission taxes on collateral policy. First, we evaluate how the long run climate policy *stance* affects macroeconomic aggregates that are relevant in the context of collateral frameworks. By reducing the return on capital, climate policy bears implications for firms' credit demand in the long run. Second, we study the short run implications of *unanticipated changes* to climate policy in the form of emission tax shocks. A sudden tax increase reduces emissions but is also a source of macro-financial risk by increasing default risk in the production sector. This suggests that climate policy affects both the quantity and quality of available collateral. So far, there is little knowledge about the implications of these effects for optimal central bank collateral policies.

To evaluate how the net zero transition affects the central bank collateral framework and its optimal design, we set up and quantify a DSGE model. There are two important frictions in the real economy. First, non-financial firms generate socially harmful emissions as a byproduct of production and emissions are taxed by the government. Second, firms can reduce their emission intensity by engaging in costly emission abatement activities (Heutel, 2012). The optimal emission intensity decreases in the tax, which reduces the return on investment since firms need to pay a higher abatement cost. While the presence of socially harmful emissions motivates climate policy in the first place, the second friction - costly emission abatement - implies that ambitious climate policy itself affects macroeconomic outcomes.

The third key friction is related to firm financing in the production sector. Firms are managed by impatient firm owners. They finance their investment by issuing equity and or by raising debt from banks, which are owned by households. Debt financing incentives arise because firm managers are more impatient than households, i.e. their time discount rate is higher than the deposit rate at which banks can finance themselves. Firms receive uninsurable idiosyncratic productivity shocks and default on their debt if revenues fall short of the debt repayment obligation (Gomes et al., 2016). Firm revenues are lost in this case, i.e. corporate default exerts a deadweight loss. The optimal debt issuance balances the benefits of debt issuance, i.e. the possibility to front-load dividend payouts, against expected default cost.

Debt is held and priced by banks, which are subject to uninsurable liquidity shocks. This is the fourth key friction in the model. Banks settle liquidity shocks by borrowing from the central bank and have to pledge debt securities as collateral (Bianchi and Bigio, 2022; De Fiore et al., 2024). The collateral eligibility of corporate debt gives rise to a collateral premium in the

corporate debt price. The collateral premium depends on the extent of banks' liquidity needs and on the central bank collateral policy. While banks benefit from a lenient collateral policy accepting risky debt securities is costly for the central bank, either because operating a collateral management system is costly (Bindseil and Papadia, 2006; Hall and Reis, 2015) or because they are averse to reporting losses from credit operations (Goncharov et al., 2023).

The presence of corporate default risk and demand for central bank liquidity gives rise to a collateral policy trade-off in this model.¹ A lenient collateral policy enables banks to smooth liquidity shocks more easily. At the same time, this policy directly increases the costs of being exposed to risky collateral for the central bank. Furthermore, the collateral premium makes debt issuance, investment and risk-taking more attractive for non-financial firms, see also the discussion in Giovanardi et al. (2023). An appealing feature of the framework is that it allows for behavioral responses to changes in collateral policy by banks and firms that go beyond these two rather mechanical effects. Both, the *quantity* of collateral available to banks and its *quality* are fully endogenous. A lenient collateral policy increases investment and debt issuance in the non-financial sector, consistent with the empirical literature on the transmission of collateral policy, see for example Pelizzon et al. (2024) or Hüttl and Kaldorf (2024). This has a positive effect on GDP and further increases available collateral. At the same time, lenient collateral policy induces risk-taking in the non-financial sector, see Bekkum et al. (2018), which increases resource losses from corporate default and reinforces the collateral management cost incurred by the central bank. The optimal collateral policy balances the liquidity benefits to banks with the downsides of additional default risk, fully taking all endogenous responses into account.

The model is calibrated to match important features of the corporate debt market, such as the leverage ratio and default frequency in the non-financial sector as well as the collateral premium. The parameterization of the climate block follows the recent E-DSGE literature while the parameters governing the real side are in line with the real business cycle literature. Central banks' collateral management costs are set in a way that the empirically observed collateral framework is optimal from a utilitarian welfare perspective in the initial long run equilibrium with low emission taxes. While the basic collateral policy trade-off is largely shaped by the two key financial frictions in the banking and non-financial sector, the two environmental frictions affect how this trade-off is resolved optimally.

Equipped with the calibrated model, we first consider the implications of emission taxes for the macroeconomy and for the basic collateral policy trade-off in the long run. Higher emission taxes reduce socially harmful emissions, which lowers emission damages. However, they also reduce the return on capital and induce divestment. Crucially, default risk is not affected by the climate policy stance, as emission taxes do not affect firms' incentives to issue debt or equity. This has two important consequences: First, divestment goes hand in hand with a reduction in debt issuance, such that the *quantity* of collateral shrinks. Second, the *quality* of collateral does not change. As a result, optimal collateral policy is structurally more lenient if the climate

¹Tabakis and Tamura (2013) provide institutional background about the trade-offs faced by the Eurosystem when setting its collateral framework. Further details on the assessment of credit risks in the Eurosystem collateral framework are available under [this link](#).

policy stance tightens. The effects are quantitatively relevant: in the baseline calibration, the optimal amount of eligible assets increases by around 20 percent.

As a second step, we consider the case of transition risk, which turns out to have very different implications for optimal collateral policy. How does the realization of transition risk affect the basic collateral policy trade-off? We show that unanticipated shocks to the emission tax increase the corporate default rate on impact. While they reduce the revenues firms generate and depress investment incentives, they do not affect firms' repayment obligations from legacy debt. Consequently, the leverage chosen in the previous period is unsustainable for a larger share of firms. The default rate rises by around 5 basis points for a 10\$/tCO₂ emission tax shock. As noted before, climate policy does not affect firms' incentives to issue debt or equity. Once firms are allowed to adjust their leverage by reducing debt issuance, the default rate reverts back to its lower long run level. At the same time, higher emission taxes still reduce investment, such that available collateral declines temporarily.

Different to the long run effects, the key collateral policy trade-off is now affected by two opposing channels. If the central bank would keep the collateral framework constant, this increases their collateral management costs. At the same time, the deleveraging pressures in the non-financial sector elevate banks' liquidity management cost. How this trade-off is resolved, depends on the shock persistence. Under i.i.d. tax shocks, the first effect clearly dominates and it is optimal to tighten collateral policy in response to a sudden increase in taxes. However, it turns out that, under any reasonable persistence of tax shocks, the optimal collateral policy adjustment in response to an emission tax shock is restrictive. For example, in our baseline calibration, it is optimal to increase collateral haircuts uniformly by 1.5 percentage points in response to a 10\$/tCO₂ emission tax shock. Reassuringly, the optimal collateral policy results are robust to various modifications of the climate block and to reasonable variations in the targeted collateral premium.

Related Literature The interactions between central bank policies and the net zero transition has received considerable attention by policymakers and the academic literature. One strand of the literature focuses on the potential role of (unconventional) monetary policy to initiate or at least support the transition to emission-free technologies. Ferrari and Nispi Landi (2023) show that green QE has a quantitatively negligible effect on aggregate emissions. Similarly, Giovanardi et al. (2023) find that green-tilted collateral policy has only a tiny effect on investment in the green sector and at the same time induces adverse side effects on financial markets. Complementary to these results, there is a rapidly evolving literature discussing how the net zero transition affects macro-financial aggregates that are relevant to the conduct of central bank policies. Berthold et al. (2023) document that emission tax shocks tighten aggregate financial conditions. Carattini et al. (2023) propose a two-sector E-DSGE model to study how sudden increases in emission taxes can lead to socially inefficient asset stranding and a credit crunch that contracts aggregate economic activity. Annicchiarico et al. (2023) study macroprudential policies if climate policy acts as an amplifier of business cycles. By contrast, McKibbin et al. (2020); Nakov and Thomas (2023); Fornaro et al. (2024); Economides and Xepapadeas (2025);

and Giovanardi and Kaldorf (2024) study different aspects of climate change and conventional monetary policy in comparable E-DSGE models without considering collateral frameworks.

This paper also contributes to the literature on central bank collateral policy and interbank markets. Koulischer and Struyven (2014) propose an analytically tractable model where banks have a funding choice between the private market and collateralized central bank borrowing. In their model, adverse shocks to the quantity or quality of collateral impair the transmission of conventional monetary policy. In a related setup, Cassola and Koulischer (2019) show how changes to the central bank collateral framework affect banks' funding cost. By contrast, Ashcraft et al. (2011) demonstrate that central bank collateral policy also affects the prices of eligible assets. Bianchi and Bigio (2022) and De Fiore et al. (2024) study the macroeconomic implications of frictions on the interbank market. While the model used in this paper shares certain frictions, such as uninsurable bank liquidity shocks, with these papers, we focus on the role of debt securities issued by the non-financial sector to collateralize central bank borrowing.

Outline The paper is structured as follows. Section 2 introduces a DSGE model with financial and environmental frictions. Its calibration is presented in Section 3. We outline the effect of climate policy on macroeconomic aggregates and optimal collateral policy in the long run in Section 4. Section 5 considers the impact of carbon tax shocks on optimal collateral policy. Section 6 demonstrates that the results are robust to varying key parameters governing financial and environmental frictions. Section 7 concludes.

2 Model

Time is discrete and denoted by $t = 0, 1, \dots$. The representative household consumes the homogeneous final good, supplies labor to final good firms and saves in the form of bank deposits. Banks invest these deposits into defaultable debt securities issued by manufacturing firms, who use debt and equity to finance their investment. Manufacturing firms sell their output to competitive final good producers.

There are three key frictions in this model. First, banks are subject to uninsurable idiosyncratic liquidity shocks that they settle with the central bank using debt securities as collateral, giving rise to collateral premia. Second, firms are subject to uninsurable productivity shocks, such that their debt securities are subject to default risk. These two frictions shape the quantity and quality of collateral and, thereby, the collateral policy trade-off. Third, manufacturing firms generate socially harmful emissions as a by-product of their production. Emissions are taxed by the fiscal authority at a potentially time-varying rate. Taxes affect the collateral policy trade-off through investment, debt issuance and default risk in the non-financial sector. We now describe each agent's decision problem.

Households Households choose consumption c_t , labor supply n_t and deposits d_t to maximize lifetime utility. Deposits pay the real interest rate i_t , while the real wage is denoted by w_t . The parameter β is the time-invariant discount factor. The maximization problem can be written

recursively:

$$V_t = \max_{c_t, n_t} \log(c_t) - \zeta \frac{n_t^{1+\gamma}}{1+\gamma} + \beta \mathbb{E}_t [V_{t+1}] \quad (1)$$

$$\text{s.t. } c_t + d_{t+1} = w_t n_t + (1 + i_{t-1})d_t + \text{div}_t + T_t ,$$

where div_t collects dividends from banks and firms. T_t is a lump sum transfer from the government. Solving the maximization problem (1), we obtain an optimality condition for labor supply $c_t^{-1} w_t = \zeta n_t^\gamma$ and the Euler equation for deposits

$$1 = \mathbb{E}_t [\Lambda_{t,t+1}(1 + i_t)] , \quad (2)$$

where $\Lambda_{t,t+1} \equiv \beta \frac{c_{t+1}^{-1}}{c_t^{-1}}$ is the household's stochastic discount factor (sdf).

Banks Banks collect deposits from households and acquire debt securities b_t from manufacturing firms. The payoff per unit of debt $\mathcal{R}_t$ depends on firm decisions, in particular on corporate default, described below. The balance sheet identity is

$$(1 + i_{t-1})d_t = \mathcal{R}_t b_t . \quad (3)$$

In the first sub-period period, banks draw uninsurable liquidity shocks that arise from (unmodeled) deposit withdrawals (Bianchi and Bigio, 2022) or similar demand for intra-period liquidity. As the focus lies on central bank collateral frameworks, we assume that the banking sector is unable to settle all idiosyncratic liquidity shocks on the interbank market, for example due to rationing of individual institutions. The collateral value of each individual banks' debt portfolio is given by

$$\bar{b}_t = (1 - \phi_t) b_t \mathcal{R}_t . \quad (4)$$

Here, b_t is debt outstanding that firms chose in the previous period and $\mathcal{R}_t$ is the realized payoff per unit of debt, described below. ϕ_t is the central bank collateral parameter. Since intra-period loans are extended before firm-specific productivity shocks realize in the second sub-period, the collateral value of debt securities is uncertain and the central bank is exposed to losses from liquidating risky collateral in the case of counterparty default. To cushion against such losses, which we describe in detail below, central banks in practice only accept sufficiently safe assets as collateral and also applies valuation haircuts to them. The policy parameter ϕ_t encompasses eligibility criteria and valuation haircuts in a tractable fashion. Specifically, $\phi_t = 1$ would correspond to the limit case where all assets receive a haircut of 100%, i.e. are ineligible, while $\phi_t = 0$ is the most lenient collateral framework possible under which banks could pledge their entire portfolio, valued at the expected payoff $\mathcal{R}_t$.

We assume that the share $\Pi \in (0, 1]$ of banks draw a liquidity deficit. Conditional on drawing a deficit, the demand of individual banks for central bank funding λ_t is exponentially distributed

with rate parameter l_1 .² If the collateral value of a bank's portfolio falls short of the demand for central bank funding, banks need to acquire funds from other, more expensive sources, for example from issuing equity and bank bonds, borrowing on the unsecured money market, or by attracting new deposits. We summarize the cost associated with all these (unmodelled) funding sources in the parameter ζ .

The expected cost from liquidity management are hence obtained from evaluating the cdf of the exponential distribution at $\bar{b}_t$ and multiplying this by the probability of receiving the negative liquidity shock Π and the cost of acquiring funding ζ : $\mathcal{L}(\bar{b}_t) = \Pi \cdot \zeta \cdot \exp\{-l_1 \bar{b}_t\}$. Since our calibration strategy is tailored to macroeconomic moments, the parameter combination $\Pi \cdot \zeta$ is observationally equivalent and we re-parameterize the cost function as follows:

$$\mathcal{L}(\bar{b}_t) = \frac{l_0}{l_1} \exp\{-l_1 \bar{b}_t\}. \quad (5)$$

Bank dividends in period t are reduced by the liquidity management cost:

$$\text{div}_t = \mathcal{R}_t b_t - (1 + i_{t-1})d_t + d_{t+1} - q(\bar{m}_{t+1})b_{t+1} - \mathcal{L}(\bar{b}_t) \quad (6)$$

We follow Cúrdia and Woodford (2011) and assume that banks can invest into debt securities as long as they are able to repay depositors in expectations. Solving the shareholder value maximization problem then yields a break-even condition for debt securities:

$$q(\bar{m}_{t+1}) = \frac{\mathbb{E}_t[\mathcal{R}_{t+1}(1 + (1 - \phi_t)\Omega_{t+1})]}{1 + i_t}. \quad (7)$$

The expression $\Omega_t \equiv -\frac{\partial \mathcal{L}_t}{\partial b_t}$ is a wedge in the debt pricing condition that is associated with the collateral benefits that debt securities provide to banks.

Manufacturing Firms: Technology Manufacturing firms use capital to produce the homogeneous intermediate good z_t which they sell to final goods producers, taken as given the price p_t^Z . Their production function is linear and subject to idiosyncratic productivity shocks $z_t = m_t k_t$. Manufacturing firms generate socially harmful emissions $e_t = (1 - \nu_t)z_t$ as a by-product of the production process. Emissions are taxed at a potentially time-varying rate τ_t and firms can reduce their emission intensity ν_t by paying a cost that is proportional to production z_t (Heutel, 2012). The emission tax bill in period t is given by $\tau_t(1 - \nu_t)z_t$. Abatement costs are specified as a simple polynomial $\frac{a_1}{a_2+1}\nu_t^{a_2+1}z_t$ with $a_1, a_2 > 0$. Firms choose their abatement effort to minimize the sum of the emission tax bill and abatement costs. Solving this static cost minimization problem yields the optimal abatement effort:

$$\nu_t^* = \left(\frac{\tau_t}{a_1}\right)^{\frac{1}{a_2}}. \quad (8)$$

²Note that our calibration is annual, such that central bank lending in this model also includes long term refinancing operations. Funding needs can vary drastically between banks over such horizons and it is reasonable to assume that some banks in fact exhaust their collateral over such an horizon. This would arguably be less likely over a daily horizon.

We define the climate policy *compliance cost* ξ_t per unit of output as:

$$\xi_t \equiv \tau_t \left(1 - \left(\frac{\tau_t}{a_1} \right)^{\frac{1}{a_2}} \right) + \frac{a_1}{a_2 + 1} \left(\frac{\tau_t}{a_1} \right)^{\frac{a_2+1}{a_2}}. \quad (9)$$

Note that under full abatement ($\tau_t = a_1$), the emission tax in the first part of equation (9) is zero, but firms still pay the abatement cost reflected in the second part of equation (9). The total abatement cost are a resource loss that enters the market clearing condition for final goods:

$$\mathcal{A}_t = \frac{a_1}{a_2 + 1} \left(\frac{\tau_t}{a_1} \right)^{\frac{a_2+1}{a_2}} z_t. \quad (10)$$

The pre-unit compliance cost (9) and the abatement cost (10) both increase in the emission tax rate. The expected compliance cost in future periods directly enter the first-order condition for capital, as we shall see below.

Manufacturing Firms: Financial Frictions Capital is financed through a combination of equity, modelled as a transfer from households, and debt securities b_t to banks. We assume that firms are operated by impatient firm-owners with discount factor $\tilde{\beta} < \beta$, i.e. they have a preference to front-load dividend payments. This generates a motive to raise additional funds from banks via debt issuance. As customary in the literature, aggregate risk is shared perfectly between the representative household and firm owners, such that firm-owners sdf is given by $\tilde{\Lambda}_{t,t+1} \equiv \tilde{\beta} \frac{c_{t+1}^{-1}}{c_t^{-1}}$.

Debt securities are long term: a share χ of debt outstanding matures each period, while the remainder $1 - \chi$ is rolled over at the current market price. After the realization of the idiosyncratic productivity shock, an individual firm defaults if revenues $p_t^Z m_t k_t$ fall short of the repayment obligation. The threshold productivity shock realization is denoted by

$$\bar{m}_t \equiv \frac{\chi b_t}{(p_t^Z - \xi_t) z_t} \quad (11)$$

We assume that m_t is log-normally distributed with standard deviation ς_m and mean $\mu_m = -\frac{1}{2}\varsigma_m^2$ to ensure that idiosyncratic productivity aggregates to one in all periods. Integrating out the idiosyncratic productivity shock, the expected payoff per unit of debt in the next period is given by

$$\mathcal{R}_{t+1} = \chi(1 - F(\bar{m}_{t+1})) + (1 - \chi)q(\bar{m}_{t+2}), \quad (12)$$

where $F(\bar{m}_{t+1})$ is the cdf of the log normal distribution. Note that $\mathcal{R}_{t+1}$ is still a random variable due to the presence of aggregate risk. Following Gomes et al. (2016), we assume that firms are restructured immediately, such that the credit status of individual firms does not enter as a state variable and the pricing of the non-maturing share of debt is unaffected. Together with the assumption that aggregate risk is shared perfectly between firm-owners and households,

this allows us to aggregate firms into a representative agent and also implies that the firm maximization problem reduces to a two-period problem:

$$\begin{aligned} \max_{k_{t+1}, b_{t+1}, \bar{m}_{t+1}} & -\psi_t k_{t+1} + q(\bar{m}_{t+1}) \left(b_{t+1} - (1 - \chi) b_t \right) \\ & + \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \cdot \left\{ \int_{\bar{m}_{t+1}}^{\infty} (p_{t+1}^Z - \xi_{t+1}) m k_{t+1} - \chi b_{t+1} dF(m) \right. \right. \\ & \left. \left. + \psi_{t+1} (1 - \delta_K) k_{t+1} + q(\bar{m}_{t+2}) \left(b_{t+2} - (1 - \chi) b_{t+1} \right) \right\} \right], \end{aligned}$$

subject to the default threshold (11) and the pricing condition on debt securities (7). Importantly, firms take into account that a change in their debt issuance affect the debt price through the default probability in future periods. Denoting the Lagrange-multiplier on the default threshold (11) by μ_t , the first-order condition for debt issuance is

$$q(\bar{m}_{t+1}) - \mu_t \frac{\bar{m}_{t+1}}{b_{t+1}} = \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \left\{ \chi (1 - F(\bar{m}_{t+1})) + (1 - \chi) q(\bar{m}_{t+2}) \right\} \right], \quad (13)$$

optimal investment is given by

$$\psi_t - \mu_t \frac{\bar{m}_{t+1}}{k_{t+1}} = \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \left\{ \psi_{t+1} (1 - \delta_K) + (p_{t+1}^Z - \xi_{t+1}) (1 - G(\bar{m}_{t+1})) \right\} \right], \quad (14)$$

while risk choice $\bar{m}_{t+1}$ solves

$$-\mu_t - q'(\bar{m}_{t+1}) \left(b_{t+1} - (1 - \chi) b_t \right) = \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \left\{ \left(b_{t+2} - (1 - \chi) b_{t+1} \right) q'(\bar{m}_{t+2}) \frac{\partial \bar{m}_{t+2}}{\partial \bar{m}_{t+1}} \right\} \right]. \quad (15)$$

The derivative of banks pricing condition of debt securities (7) with respect to the risk choice obtains from plugging in the debt payoff (12) into equation (7) and reads

$$q'(\bar{m}_{t+1}) = \mathbb{E}_t \left[-\chi F'(\bar{m}_{t+1}) + (1 - \chi) \frac{\partial \bar{m}_{t+2}}{\partial \bar{m}_{t+1}} q'(\bar{m}_{t+2}) \right] \frac{1 + (1 - \phi_t) \Omega_t}{1 + i_t}. \quad (16)$$

Note that the risk choice in the following period $\bar{m}_{t+2}$ is relevant for the pricing of debt and for today's risk choice because debt is long term. The slope of the policy function for risk $\frac{\partial \bar{m}_{t+2}}{\partial \bar{m}_{t+1}}$ depends on the unknown policy function for the risk choice (Gomes et al., 2016). We solve for the slope by adding an additional equilibrium condition. Re-arranging (13) for the multiplier μ_t ,

$$\mu_t = \frac{b_{t+1}}{\bar{m}_{t+1}} q(\bar{m}_{t+1}) - \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \frac{b_{t+1}}{\bar{m}_{t+1}} \left\{ \chi (1 - F(\bar{m}_{t+1})) + (1 - \chi) q(\bar{m}_{t+2}) \right\} \right],$$

and further differentiating with respect to $\bar{m}_{t+1}$ yields

$$q(\bar{m}_{t+1}) + \bar{m}_{t+1} q'(\bar{m}_{t+1}) = \mathbb{E}_t \left[\tilde{\Lambda}_{t+1} \left\{ \chi (1 - F(\bar{m}_{t+1})) + (1 - \chi) \left(q(\bar{m}_{t+2}) + q'(\bar{m}_{t+2}) \bar{m}_{t+1} \frac{\partial \bar{m}_{t+2}}{\partial \bar{m}_{t+1}} \right) \right\} \right] \quad (17)$$

which together with (15) determines the multiplier on the default threshold (11) in the maximization problem, and the slope of the policy function for default threshold. The aggregate

losses from corporate default are given by

$$\mathcal{F}_t = \int_0^{\bar{m}_t} m_t z_t dF(m_t) , \quad (18)$$

i.e. current production by intermediate good firms, weighted by the aggregate productivity of all defaulting firms. This term directly enters the resource constraint as default cost.³

Final Good and Capital Good Firms There is a continuum of perfectly competitive final good producers that combine labor and intermediate goods in using a Cobb-Douglas technology, subject to aggregate productivity shocks

$$y_t = A_t \exp\{-\Psi_E e_t\} z_t^\theta n_t^{1-\theta} . \quad (19)$$

Here, A_t is exogenous TFP and the term $\exp\{-\Psi_E e_t\}$ reflects damages from socially harmful emissions, described below. Denoting the intermediate good price by p_t^Z and normalizing the final good price to one, the first order conditions are standard:

$$w_t = (1 - \theta) \frac{n_t}{y_t} , \quad (20)$$

$$p_t^Z = \theta \frac{z_t}{y_t} . \quad (21)$$

The capital good that intermediate firms acquire to build up their capital stock is produced by a representative investment good firms. The capital price is denoted by ψ_t . Capital good firms convert $(1 + \frac{\Psi_I}{2}(\frac{i_t}{i_{t-1}}))$ units of the final good into one unit of the investment good, such that the profit maximization problem

$$\max_{\{i_t^s\}_{t=0}^\infty} \mathbb{E}_0 \left[\sum_{t=0}^\infty \beta^t \frac{c_t^{-\gamma_C}}{c_0^{-\gamma_C}} \left\{ \psi_t i_t - \left(1 + \frac{\Psi_I}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 \right) i_t \right\} \right]$$

yields the first-order condition linking equilibrium investment to the price of capital:

$$\psi_t = 1 + \frac{\Psi_I}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 + \Psi_I \left(\frac{i_t}{i_{t-1}} - 1 \right) \frac{i_t}{i_{t-1}} - \mathbb{E}_t \left[\Lambda_{t,t+1} \Psi_I \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] . \quad (22)$$

Public Policy and Closing the Model Emission taxes are set exogenously and we allow for some degree of policy uncertainty as source of transition risk. This is operationalized by assuming that they follow an AR(1) process:

$$\tau_t = (1 - \rho_\tau) \tau^{SS} + \rho_\tau \tau_{t-1} + \sigma_\tau \epsilon_t^\tau . \quad (23)$$

³We implicitly assume that final good producers receive $\int_0^\infty m_t dF(m_t) z_t = z_t$ goods since the idiosyncratic productivity integrates to one by assumption. Alternatively, one could assume that intermediate good supply of defaulters is zero and assume away direct default costs entering the final goods market clearing condition. This would imply a very similar resource losses but introduce a slightly more cumbersome market clearing condition for intermediate goods.

The persistence parameter in the emission tax shock is set to $\rho_\tau = 0.8$, which is the same persistence as the TFP shock. To calibrate the model, we set the standard deviation to $\sigma_\tau = 0.00013$, such that a one standard deviation shock corresponds to 1\$/tCO2. This value corresponds to the empirical estimate by Berthold et al. (2023).

The central bank extends intra-period loans at a zero interest rate against collateral. Following Bindseil and Papadia (2006) and Hall and Reis (2015), accepting risky debt securities as collateral is costly for at least two reasons. First, central banks need to operate a credit risk management system, which is not directly related to their typical monetary policy activities. Second, in the case of a counterparty default, the central bank needs to liquidate the pledged collateral, potentially at a loss. We assume that the cost $\mathcal{C}_t$ are quadratic in the amount of eligible collateral $\bar{b}_t$ and the realized default frequency of the pledged securities $F(\bar{m}_t)$:

$$\mathcal{C}_t = c_0 \cdot (\mathcal{B}_t)^2 \quad \text{where} \quad \mathcal{B}_t \equiv \bar{b}_t \cdot F(\bar{m}_t). \quad (24)$$

The central bank exposure to losses from intra-period loans to banks $\mathcal{B}_t$ is defined as the potential size of loans bank can obtain, weighted by the corporate default rate. We focus on the contemporaneous default frequency $F(\bar{m}_t)$ because central bank losses realize in the second sub-period, when the intra-period central bank loans mature. The scale c_0 in the cost function is set such that the long run collateral policy ϕ^{SS} is optimal from a utilitarian welfare perspective, following the approach in Giovanardi et al. (2023).⁴ The collateral framework can respond to emission taxes, which we operationalize this by a simple rule in the spirit of Schmitt-Grohé and Uribe (2007):

$$\phi_t = \phi^{SS} + \varphi_\tau \cdot \mathcal{S} \cdot (\tau_t - \tau^{SS}). \quad (25)$$

Here, the auxiliary constant $\mathcal{S}$ converts model emission taxes, which are in abstract units, into US dollars per tonne of carbon dioxide (\$/tCO2). We describe this conversion factor in the calibration section. The model is closed by assuming that the central bank rebates its net profits to households in lump sum fashion. Market clearing for final goods requires:

$$y_t = c_t + i_t \cdot \left(1 + \frac{\Psi_I}{2} \left(\frac{i_t}{i_{t-1}} - 1\right)^2\right) + \mathcal{A}_t + \mathcal{C}_t + \mathcal{F}_t + \mathcal{L}_t. \quad (26)$$

Abatement costs, collateral management costs incurred by the central bank, resources losses from corporate default, and liquidity management costs incurred by banks all reduce household consumption. Lastly, exogenous TFP follows an AR(1) process in logs:

$$\log(A_t) = \rho_A \log(A_{t-1}) + \sigma_A \epsilon_t^A. \quad (27)$$

⁴If the liquidity shock can be partially settled on the interbank market, the central banks' exposure would be smaller than $\bar{b}_t$. Under this interpretation, we would need to re-calibrate c_0 . As the empirical strategy is based on eligibility premia and aggregate corporate debt supply, this is observationally equivalent to the setting where all liquidity shocks are settled with the central bank directly.

3 Calibration

Each model period corresponds to one year. Parameters governing households and the production technology are set to conventional values used in the real business cycle literature. To calibrate parameters concerning financial frictions, in particular debt issuance, default risk and the collateral premium, we follow recent estimates from the empirical literature. Lastly, parameters in the climate block are in line with smaller scale environmental DSGE models. We describe each group of parameters in turn. The full parameterization is summarized in Table 1.

Parameter	Value	Source
<i>Households</i>		
Household discount factor β	0.99	Standard
Labor supply curvature γ	1	Frisch elasticity of one
Labor supply weight ζ	8.5	$n^{SS} = 0.33$
<i>Technology</i>		
Cobb-Douglas coefficient α	1/3	Standard
Capital depreciation rate δ_K	0.1	Standard
Inv. adj. parameter Ψ_I	0.5	Standard
TFP persistence ρ_A	0.8	Standard
TFP shock vol σ_A	0.01	Standard
Abatement cost parameter a_0	0.03	Full Abatement Tax 220\$/tCO2
Abatement cost parameter a_0	1.6	Heutel (2012)
Emission damage parameter Ψ_E	0.03	5% Damage/GDP ratio
Emission tax level τ^{SS}	0.003	23\$/tCO2 global average tax
Emission tax persistence ρ_τ	0.8	Standard
Emission tax shock vol σ_τ	0.00013	Berthold et al. (2023)
<i>Financial Markets</i>		
Share of maturing debt χ	0.2	Five year maturity
Firm-owner discount factor $\tilde{\beta}$	0.975	Target: Firm leverage 35%
St. dev. firm risk ς_m	0.2	Target: Firm default rate 0.5%
Liquidity management cost scale l_0	0.004	Collateral premium 12bps
Liquidity management cost curvature l_1	1	Normalization
Long run collateral parameter ϕ^{SS}	0.65	Eurosystem data
Collateral default cost scale c_0	330	Optimality of ϕ^{SS}

Table 1: Baseline Calibration

Households and Production Sector In line with our annual calibration, we set $\beta = 0.99$, implying a real risk-free rate of 1% p.a. The Frisch elasticity of labor supply γ is set to its conventional value of one, while the weight on labor supply disutility $\zeta = 9$ is set such that it implies a steady state labor supply of $n^{SS} = 0.33$. The depreciation rate of physical capital $\delta_K = 0.1$, investment adjustment cost parameter $\Psi_I = 0.5$ and the Cobb-Douglas coefficient in the final goods production $\theta = 1/3$ are set to standard values used in the real business cycle literature.

Financial Sector Setting the maturity parameter to $\chi = 0.2$ implies an average debt maturity of five years. The curvature parameter l_1 in banks' liquidity management cost is normalized to one. We inform the scale parameter $l_0 = 0.004$ using recent empirical estimates of the collateral premium. The model-implied collateral premium is defined as the yield differential between the traded debt security and a counterfactual security with the same payoffs but without the collateral premium $(1 - \phi_{t+1})\Omega_{t+1}$. Similar to the price of the traded security (7), the counterfactual security is priced using the following recursion:

$$\tilde{q}(\bar{m}_{t+1}) = \frac{\chi(1 - F(\bar{m}_{t+1})) + (1 - \chi)\tilde{q}(\bar{m}_{t+2})}{1 + i_t}. \quad (28)$$

Based on the random maturity structure of debt securities, the yield to maturity for the traded security is defined implicitly through the relationship $q(\bar{m}_{t+1}) = \frac{\chi + (1 - \chi)q(\bar{m}_{t+2})}{1 + r_t}$. Note that the yield on the traded security contains both a default risk premium and a collateral premium. Re-arranging gives $r_t = \frac{\chi}{q(\bar{m}_{t+1})} - \chi$. In a similar fashion, we obtain the yield to maturity for the counterfactual debt security $\tilde{r}_t = \frac{\chi}{\tilde{q}(\bar{m}_{t+1})} - \chi$. The collateral premium is then defined as $\tilde{r}_t - r_t$ and set to 12 basis points, in line with recent empirical work by Bekkum et al. (2018), Mésonnier et al. (2021), Pelizzon et al. (2024), or Hüttl and Kaldorf (2024). We set the firm discount factor $\tilde{\beta} = 0.975$ and the standard deviation of the idiosyncratic shock to $\varsigma_m = 0.2$ to jointly match a corporate leverage ratio of 35% and a default frequency of 0.5% per annum.

To calibrate the collateral policy parameter ϕ^{SS} one has to bear in mind that it encompasses both the share of all debt securities that banks can pledge as collateral with the central bank, and the valuation haircuts applied to them. We use publicly available macro aggregates and apply some institutional details from the Eurosystem haircut schedule to obtain a data moment for $\bar{b}_t$. Using Eurosystem data from 2015 to 2024, the ratio of total eligible assets over the balance sheet size of the banking sector corresponds to 45% on average.⁵ This would imply a collateral parameter of $\phi^{SS} = 55\%$. As there are no consolidated data on average haircuts of the eligible collateral available, we add a discretionary valuation haircut of 10%, which reduces the collateral value of banks portfolio. As of 2025, bonds issued by non-financial corporation with a residual maturity of 5-7 years and a BBB-rating receive a 15% haircut in the Eurosystem collateral framework, while bonds with a A-rating or higher get a 5% haircut. Lower rated bonds are not eligible. Investment grade corporate bonds appear to be representative of the assets accepted as collateral by the Eurosystem and they are also similar to the debt securities used in the model.⁶ This leaves us with an average collateral parameter of $\phi^{SS} = 65\%$. The scale parameter in the collateral management cost (24) is set to $c_0 = 330$, such that the collateral policy parameter is optimal from a utilitarian welfare perspective. We show in Section 6 that all results are robust to reasonable variations of ϕ^{SS} .

⁵For the size of the banking sector, we use total assets reported by all MFIs in the euro area, which is publicly available under [this link](#). Eurosystem collateral data are available under [this link](#).

⁶The haircuts applicable to different asset classes in the Eurosystem are available under [this link](#).

Moment	Model	Data	Source
Relative vol consumption $\sigma(c)/\sigma(y)$	0.83	0.96	Euro area data
Relative vol investment $\sigma(i)/\sigma(y)$	1.73	1.90	Euro area data
Correlation Consumption-GDP $\text{corr}(c, y)$	0.97	0.91	Euro area data
Correlation Investment-GDP $\text{corr}(i, y)$	0.94	0.86	Euro area data
Correlation Emissions-GDP $\text{corr}(e, y)$	0.55	0.76	Euro area data
Correlation Default-GDP $\text{corr}(F(\bar{m}_t), y)$	-0.44	-0.55	Kuehn and Schmid (2014)
Correlation Debt-GDP $\text{corr}(b_t, y_t)$	0.98	0.65	Jungherr and Schott (2022)

Table 2: Model Fit: All model-implied moments are computed based on a second-order approximation around the deterministic steady state and expressed in relative deviations from their steady state value. The long run emission tax is fixed at 23\$/tCO2 and the standard deviation of emission tax shocks is set to $\sigma_\tau = 0.0013$. All euro area data moments are based on real data, are logged and de-trended using an one-sided HP-filter with smoothing parameter of 6.25.

Climate Block Following Heutel (2012), the curvature parameter of the abatement cost function is set to $a_2 = 1.6$. The level parameter a_1 is set to 0.03 such that an emission tax of 230\$/tCO2 induces full abatement of emissions. This value is in line with Bilal and Kaenzig (2025) and Environmental Defense Fund (2021). To make the model-implied emission tax interpretable, it is converted into an emission price p^e , expressed in \$/tCO2. This price is related to the emission tax through $p^e \cdot \frac{e^{\text{world}}}{y^{\text{world}}} = \tau \cdot \frac{e^{\text{model}}}{y^{\text{model}}}$, such that we can convert the emission tax from model units into \$/tCO2 using the following relationship:

$$p^e = \frac{y^{\text{world}}/y^{\text{model}}}{e^{\text{world}}/e^{\text{model}}} \cdot \tau \equiv \mathcal{S} \cdot \tau. \quad (29)$$

The conversion rate $\mathcal{S}$ will be used in the We use the ratio of model output ($y^{\text{model}} = 0.626$ in the long run equilibrium without emission taxes) to global GDP ($y^{\text{world}} = 105$ billion USD in 2022) and model emissions ($e^{\text{model}} = 0.349$ in the long run equilibrium without emission taxes) to global emissions ($e^{\text{world}} = 37.5$ GtCO2 in 2022). The emission tax in the initial steady state is set to 23\$/tCO2, which corresponds to the global average of emission taxes and prices in emission trading schemes in 2022.

Model Fit Table 2 illustrates that the model is capable of replicating key macroeconomic aggregates that are important for the analysis of collateral policy in the context of the net zero transition. The excess volatilities of consumption and investment over GDP, as well as their correlations with GDP are in line with their data counterparts. Aggregate emissions are strongly pro-cyclical. By contrast, corporate default risk is counter-cyclical, i.e. it spikes during recessions. The debt issuance is more pro-cyclical than in the data, as the model features only two exogenous shocks (TFP and emission taxes) that are directly linked to the production side. Debt issuance closely follows investment demand, such that it has to be highly correlated with aggregate output.

4 Optimal Collateral Policy in the Long Run

Using the calibrated model, we discuss how emission taxes affect key macroeconomic aggregates and the optimal conduct of collateral policy in the long run. As a first step, we compute the long run equilibrium for the range of emission taxes between their current level of 23\$/tCO₂ up to 230\$/tCO₂. By the first-order condition for abatement, equation (8), this emission tax level induces net zero emissions. The abatement level ν_t^* implied by each tax rate is shown in the top left panel of Figure 1. The other panels of Figure 1 display key welfare relevant aggregates that shape the conduct of central bank collateral policy.

Crucially, the corporate default rate is independent of the climate policy regime. The reason for this perhaps puzzling observation is that the emission tax does not change the risk choice $\bar{m}_{t+1}$, see equation (15). In the next section, we will see that the irrelevance of climate policy for corporate default risk does not hold any longer in the presence of emission tax shocks: non-financial firms cannot immediately adjust their risk-taking which renders their indebtedness unsustainable if taxes are increased suddenly.

If firms can adjust to climate policy, the emission tax only affects the first-order condition for investment (14). Figure A.1 in Section A demonstrates that ambitious climate policy is generally accompanied by a decline in aggregate capital. As the risk choice not affected, the first-order condition for debt (13) dictates that the divestment induced by climate policy is accompanied by a proportional reduction in loan demand, such that firms indebtedness remains constant. This is shown in the bottom left panel of Figure 1. The corporate debt/GDP ratio declines from 96% to around 90%. The bottom right panel shows that this induces the collateral/GDP ratio do decline from 32% to 30%, i.e. banks have less collateral at their disposal.⁷

In Figure 2, we display how the central bank collateral parameter ϕ^{SS} affects the macroeconomy in the long run. The dashed red line refers to the case with status quo emission taxes (23\$/tCO₂), while the dotted green line reflects an emission tax of 230\$/tCO₂ consistent with net zero. The top left panel shows the amount of available collateral $\bar{b}_t = (1 - \phi_t)\mathcal{R}_t b_t$, which is directly affected by modification to the collateral parameter. For instance, setting $\phi^{SS} = 90\%$ would imply a collateral/GDP ratio of slightly less than 10%. This policy could be implemented by only allowing the 10% least risky corporate debt as collateral without applying any valuation haircut or by accepting all debt securities but attaching an haircut of 90% to all of them. In practice, the collateral framework is a complicated mapping from a set of very heterogeneous debt securities into a collateral value that could be zero if the security is ineligible. Since our macro model abstracts from such heterogeneity, ϕ^{SS} is best interpreted as a collateral policy stance encompassing both eligibility criteria and valuation haircuts.

The top right panel demonstrates that lenient collateral policy incentivizes firms to issue more debt (see also Giovanardi et al., 2023), which is also reflected in higher default rates. If collateral policy is overly lenient, higher default rates increase both the resource losses from corporate default and the collateral management cost incurred by the central bank. If it is

⁷We report the effect of long run emission taxes on GDP, investment and the different welfare relevant costs entering the resource constraint in Section A.

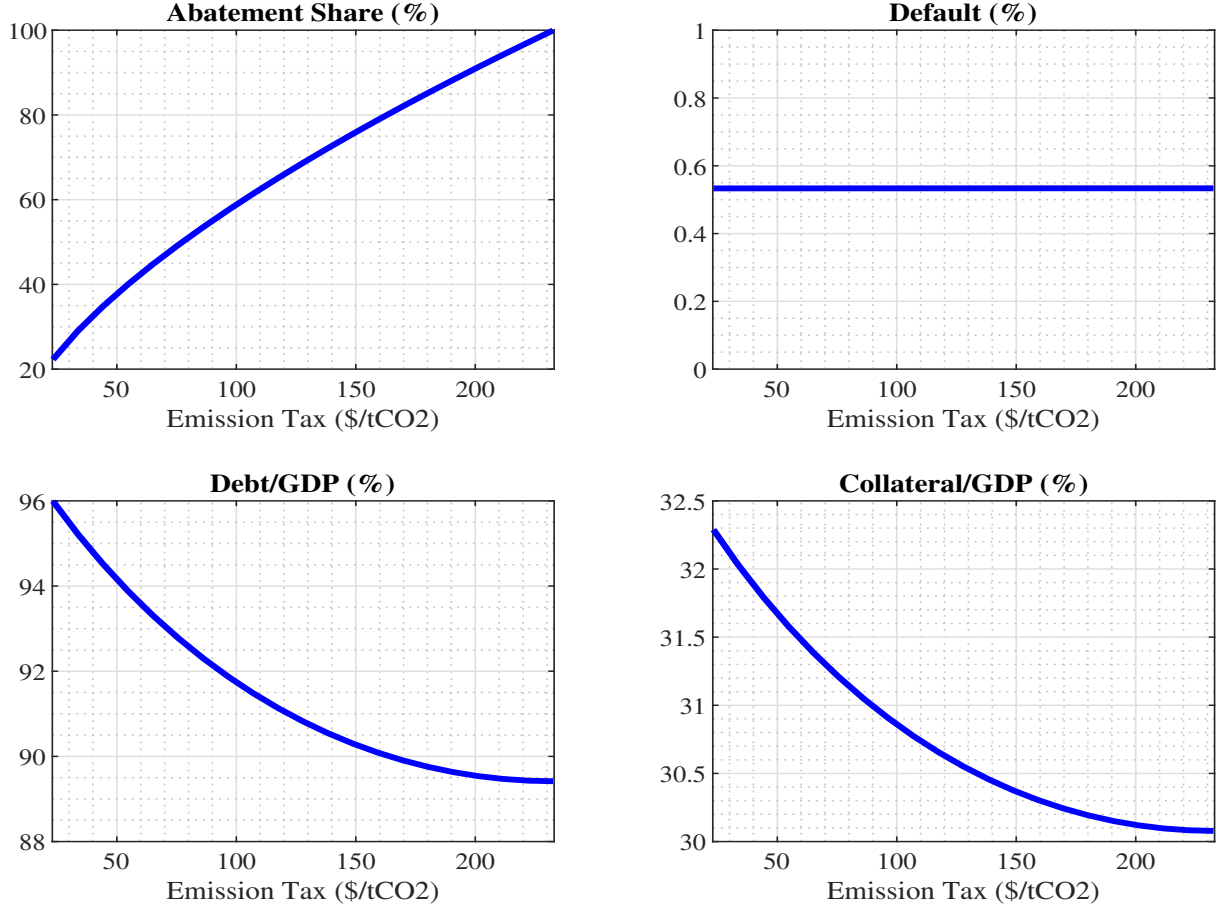


Figure 1: Long Run Effects of Emission Taxes. This figure displays the effects of long run emission taxes on key welfare relevant aggregates: Share of abated emissions ν_t^* (in %, top left), the corporate default rate $F(\bar{m}_t)$ (in percentage points, top right), the ratio of corporate debt b_t to GDP (in %, bottom left), and the ratio of collateral $\bar{b}_t$ to GDP (in %, bottom right). The emission tax on the x-axis is expressed in \$/tCO2.

overly tight, such costs are small but banks face high liquidity management costs in this case. As the dashed red line in the bottom right panel of Figure 2 shows, the baseline calibration implies that a collateral parameter of $\phi^{SS} = 65\%$ is optimal in the long run with low emission taxes. Figure A.2 in Section A shows how collateral policy affects additional macroeconomic aggregates, such as aggregate investment.

How do emission taxes affect the transmission of collateral policy to the macroeconomy and its optimal design? As we have seen in Figure 1, climate policy has no long run impact on the default rate. Hence, the quality of collateral is unchanged. At the same time, by exerting divestment and deleveraging pressure on non-financial firms, they reduce the quantity of collateral. As the bottom right panel demonstrates, the welfare optimum is reached for a smaller collateral policy parameter. We show in Section 6 that this result is robust to reasonable variations in parameters governing the climate block and financial frictions. Again, our macroeconomic framework does not provide guidance whether this should be implemented via lenient eligibility criteria or lower valuation haircuts.

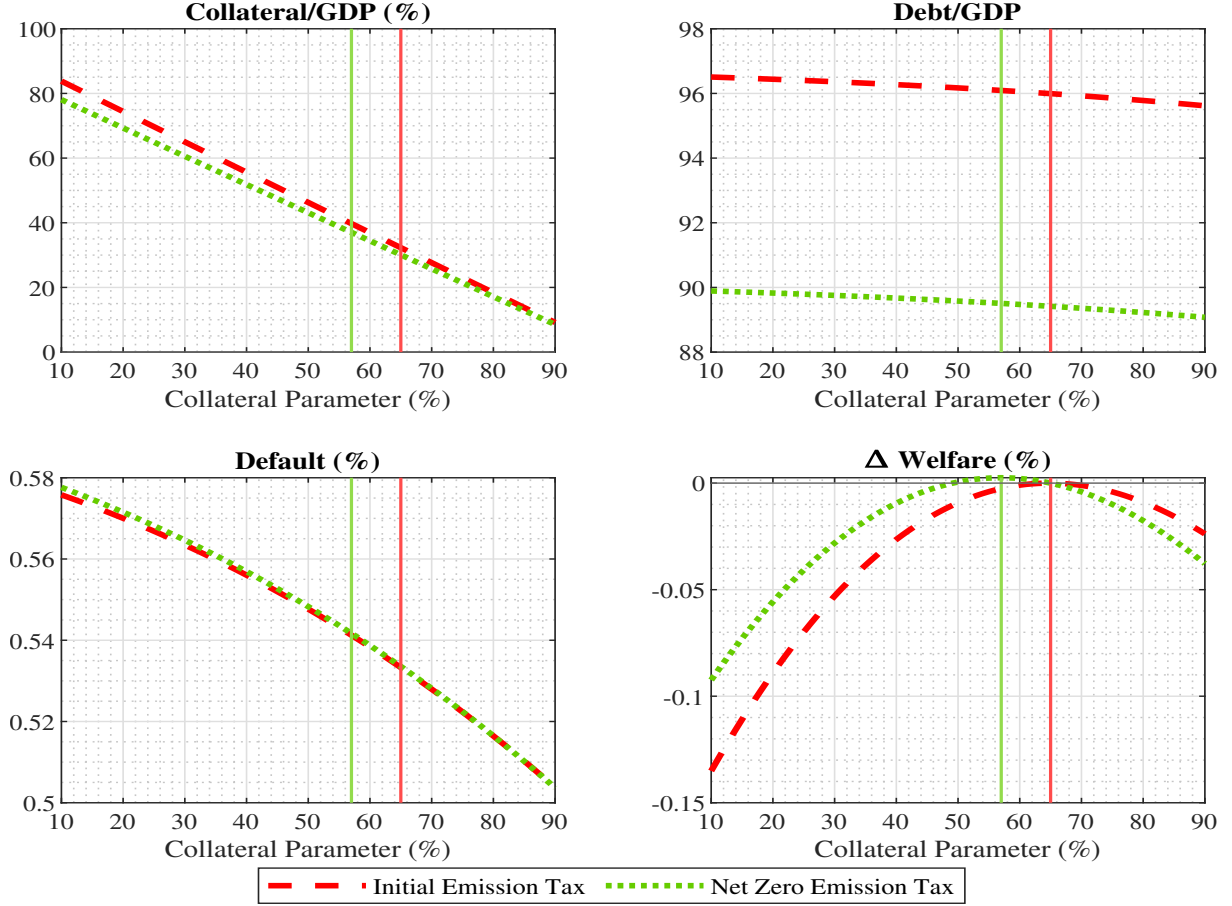


Figure 2: Optimal Long Run Collateral Policy. This figure displays the effects of varying the collateral parameter ϕ^{SS} on the ratio of collateral b_t to GDP (in %, top left), the ratio of corporate debt b_t to GDP (in %, top right), the corporate default rate $F(\bar{m}_t)$ (in percentage points, bottom left). The welfare change in the bottom right is computed relative to the long run with the baseline collateral parameter of $\phi^{SS} = 65\%$. The dashed red line refers to a 23\$/tCO2 emission tax, the dotted green line to the net zero tax of 230\$/tCO2. The collateral parameter ϕ^{SS} on the x-axis is expressed in percentage points.

5 Optimal Collateral Policy in the Short Run

Having discussed the long run implications of climate policy on the optimal conduct of collateral policy, this section turns to the impact of sudden climate policy changes. As customary in the literature, this is operationalized by subjecting emission taxes to shocks. We focus on a one standard deviation increase in emission taxes, which corresponds to an increase by 10\$/tCO2 in our calibration. Figure 3 displays the impulse responses to such an emission tax shock. The dotted blue line refers to the (constant) baseline collateral policy $\phi_t = \phi^{SS}$.

The upper panel shows that such an emission tax increase induces a substantial increase in the share of abated emissions. At the same time, this affects the default risk and debt issuance in the non-financial sector, which is key in shaping the optimal collateral policy response. The top left panel shows that the corporate default rate increases by around 4 basis points, which can be interpreted as a materialization of transition risk. At the same time, this puts immediate downward pressure on debt issuance, which declines by more than half a percentage point, and the availability of collateral, which declines by around 0.2 percentage points. In Figure A.3, we

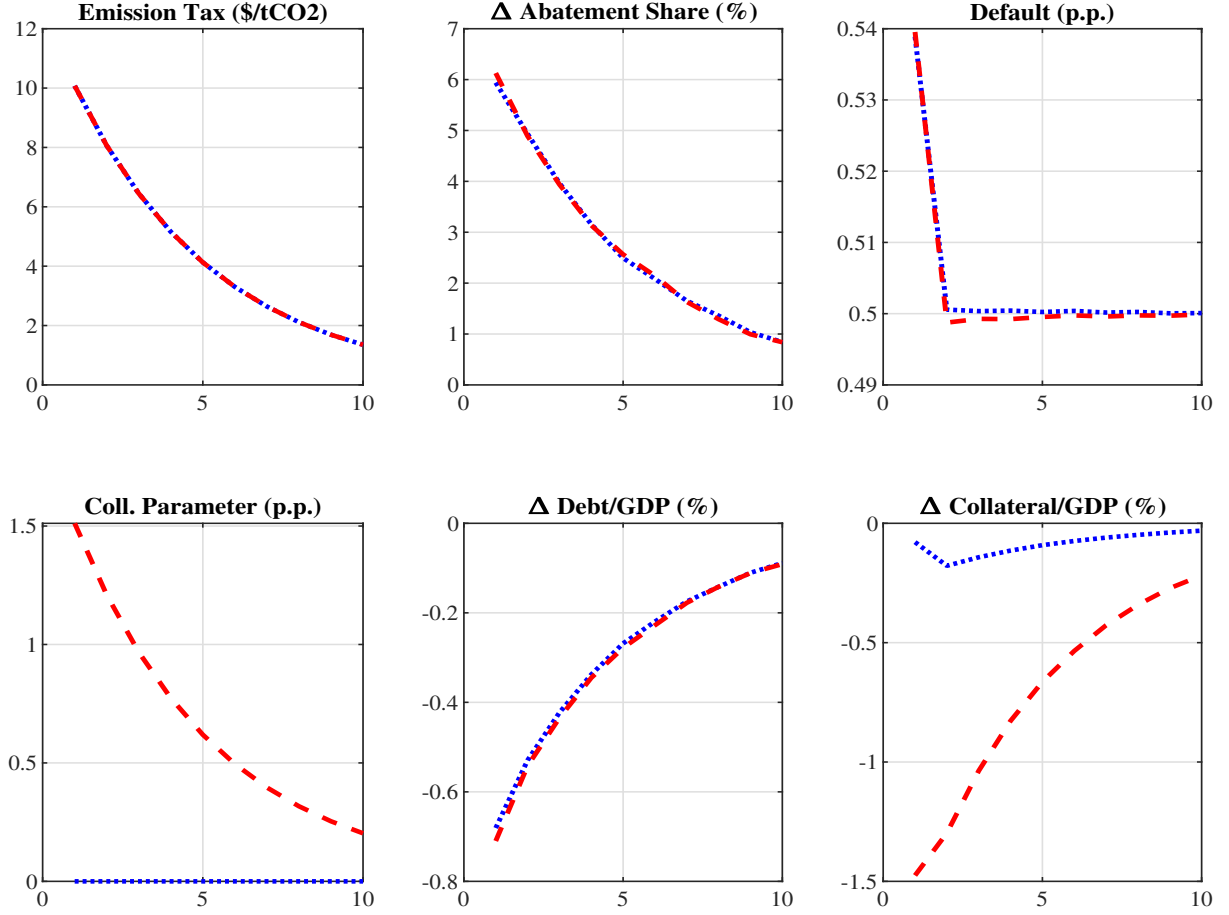


Figure 3: Emission Tax Shocks. This figure displays the effects of a 10\$/tCO₂ emission tax shock on key welfare relevant moments over time. The top row displays the percentage change of debt outstanding b_{t+1} and investment i_t , both in % deviations from steady state, as well as the default rate $F(\bar{m}_t)$ in percentage points. The bottom row shows the absolute change in the collateral parameter ϕ_t and the yields of the traded (r_t) and synthetic ($\tilde{r}_t$) debt security and the collateral premium, all expressed in basis points. The dotted blue line refers to the case of constant collateral policy, the dashed red line to optimal dynamic collateral policy.

show that this decline is driven by a reduction in aggregate investment.

How should collateral policy respond optimally to the macro-financial implications of emission tax shocks? We focus on optimal collateral policy rules, which is obtained from maximizing utilitarian welfare over φ_τ in equation (25). For this experiment, we set the tax shock standard deviation to $\sigma_\tau = 0.0013$, i.e. a one standard deviation shock corresponds to a 10\$/tCO₂ increase in emission taxes. This value is intentionally large in order to provide a reasonable upper bound on the effects of transition risk on central bank collateral policy. For the baseline persistence of $\rho_\tau = 0.8$, the collateral parameter optimally increases by 1.5 percentage points in response to a 10\$/tCO₂ shock. The dashed red line in Figure 3 displays the macro-financial implications of the optimal collateral policy adjustment after an emission tax shock. Since collateral policy does not enter the first order condition for abatement, the abatement share is does not change. In a similar vein, the increase of the default rate on impact cannot be affected by collateral policy, as the debt outstanding was chosen in the previous period.

However, by tightening collateral policy, the default rate is slightly lower in the periods

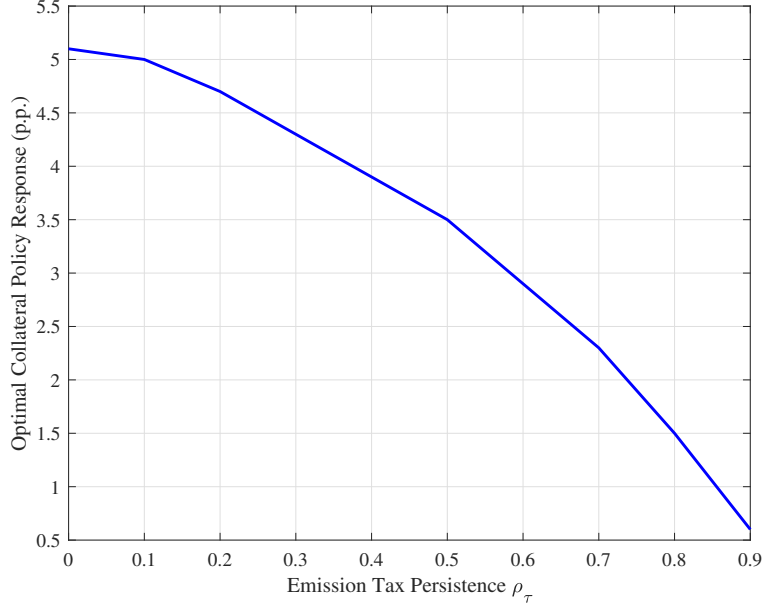


Figure 4: Optimal Collateral Policy: The Role Emission Tax Persistence. This figure displays the optimal collateral policy response φ_τ , expressed in percentage points, to a 10\$/tCO₂ emission tax shock for different degrees of persistence ρ_τ in the emission tax rate.

after the shock, as temporarily elevated haircuts provide (small) disciplining incentives for non-financial firms. This is also reflected by the slightly more pronounced decline in the debt/GDP ratio. Consequently, the collateral/GDP is considerably smaller. In Figure A.3, we show that such a contractionary policy substantially reduces the collateral management cost incurred by the central bank, which also has a substantial impact effect, as central bank loans are intra-period. This goes hand in hand with a temporary increase in banks' liquidity management cost.

Why is it optimal to tighten collateral policy in the short run? Different from the long run climate policy stance, an unexpected emission tax hike affects the corporate default rate and, by extension, the collateral management cost. These effects are not present in the long run, where the non-financial sector has time to adjust to the new climate policy regime by reducing its debt issuance. For exactly the same reason, the temporary nature of emission tax shocks implies a smaller deleveraging pressure on firms, such that collateral supply contracts comparatively less. This renders a temporary increase in ϕ_t optimal.

To further corroborate this intuition, Figure 4 shows the results of varying the persistence parameter in the emission tax process (23) and re-optimizing the response parameter φ_τ . For the case of i.i.d. shocks ($\rho_\tau = 0$), optimal collateral policy is very restrictive. The quantity of collateral hardly responds if shocks are i.i.d., since the expected emission tax in the period after the shock corresponds to the long run mean. This makes it optimal to drastically reduce collateral management cost. By contrast, for very persistent shocks, the result almost resembles the long run case. Here, the debt supply effects almost offset the short run increase in corporate default risk and it is optimal to hardly adjust the collateral framework.

6 Robustness

This section presents a set of robustness tests for the long run and short run implications of climate policy for the optimal collateral framework. First, we discuss the role of emission damages and the abatement cost function, i.e. the two key parameters in the climate block. Second, we consider a different target for the collateral premium, for which empirical estimates also vary considerably, and for the central bank cost of accepting risky collateral.

Emission Damages In the baseline model, collateral policy is more stringent in the long run when emission taxes are low. This might be optimal not only from the basic collateral policy trade-off but also from a climate policy consideration: tighter collateral policy depresses economic activity and, thereby, reduces emissions. If the elasticity of emission damages with respect to output is very large, such a reduction in economic activity can increase welfare. To alleviate the concern that the results are driven by this consideration, we set $\Psi_E = 0$ to eliminate emission damages. This clearly implies that emission taxes reduce welfare. We also recalibrate $c_0 = 430$ to ensure that the collateral parameter $\phi^{SS} = 65\%$ is optimal in the long run equilibrium with small emission taxes. Here, the optimal $\phi^{SS} = 63\%$ is still smaller in the long run equilibrium with net zero climate policy, but the effect is smaller. Similarly, the dynamic response to an emission tax shock is more aggressive in this case.

Abatement Cost There is considerable uncertainty surrounding the costs of adopting and operating emission-free technologies. In the baseline parameterization, we have used cost curves applicable to the US and Europe, which might underestimate the abatement costs at a global level. Choosing a higher abatement cost parameter $a_0 = 0.04$ implies that a larger emission tax of 310\$/tCO₂ is necessary to induce net zero emissions. This in turn exacerbates the wedge ξ_t in the first order condition for investment (14), such that capital and debt outstanding decline relatively more in the long run equilibrium with net zero climate policy. The scale parameter of collateral management cost is re-calibrated to $c_0 = 350$ in this case. To counteract the larger decline in debt issuance, the optimal long run collateral parameter under the net zero policy is slightly smaller at $\phi^{SS} = 56\%$. Conversely, the optimal collateral policy response to emission tax shocks is larger, since there is a larger impact effect on corporate default risk and the central banks' collateral management cost.

Collateral Premium In the baseline calibration, we used a collateral premium of 12bps, which is consistent with recent empirical literature. Some papers, however, find a considerably larger collateral premium. Using Chinese data, Chen et al. (2023) and Fang et al. (2025) report a collateral premium around 50bps. Thus, we show that the implications of the net zero transition on optimal collateral policy are robust to increasing the scale parameter of liquidity management costs to $l_0 = 0.008$, which implies a collateral premium of 25bps, which strikes a middle-ground between the high premium found in Chinese data and the smaller estimate based on euro area data. As before, we adjust the scale parameter of collateral default costs to render

	Optimal ϕ^{SS} Initial tax	Optimal ϕ^{SS} Net zero tax	Optimal response to 10\$/tCO2 tax φ_τ
Baseline	65	57	1.5
No climate damages $\Psi_E = 0$	65	63	2.2
Higher abatement cost $a_0 = 0.04$	65	56	1.8
Higher coll. premium $l_0 = 0.008$	65	58	1.3
Lower coll. management cost $c_0 = 300$	60	52	1.7
Higher coll. management cost $c_0 = 420$	70	63	1.4

Table 3: Robustness. This table reports the optimal long run collateral parameter in the long run equilibrium with small emission taxes (23\$/tCO2, first column) and under net zero emission taxes (second column), The third column reports the optimal collateral policy response φ_τ to a one standard deviation (10\$/tCO2) emission tax shock, see equation (25). All results are based on long simulations of a second order approximation around the deterministic steady state. The long run collateral parameter and its response to an emission tax shock is expressed in percentage points.

the long run collateral parameter $\phi^{SS} = 65\%$ optimal. This is the case for $c_0 = 550$. Under full abatement taxes, the optimal collateral parameter is $\phi^{SS} = 58\%$, which is very close to the baseline calibration. The more pronounced welfare loss of a decline in debt securities is offset by the more pronounced uptake in collateral management costs incurred by the central bank. The dynamic response $\varphi_\tau = 1.3\%$ is less pronounced than in the baseline: tightening collateral policy too much would induce a sub-optimally large increase in banks' liquidity management cost.

Collateral Management Costs In order to calibrate the long run collateral parameter ϕ^{SS} , we have used the ratio of all eligible assets in the Eurosystem over the size of the euro area banking sector. While this approach appears to be the most pragmatic way to aggregate the fairly complicated Eurosystem collateral framework into a single number that can be related to macroeconomic moments, it is naturally subject to certain biases that could go in either direction.

First, the numerator could overstate the amount of eligible assets, as we have uniformly reduced the value of assets reported by the ECB by a 5% valuation haircut. This would correspond to the valuation haircut of high quality corporate bonds with a five year residual maturity. In practice, a sizable share of eligible assets is riskier than that, either because the eligible assets are not traded on liquid markets or have a lower rating. To show that the results are robust to such a bias, we recalibrate $c_0 = 420$ which ensures that $\phi^{SS} = 70\%$ is optimal in the long run equilibrium with small emission taxes. Note that collateral premia would be slightly smaller in this situation, which has detrimental effects on debt issuance, investment and real activity. Nevertheless, the optimal collateral parameter under net zero taxes is 63%. The difference of seven percentage points is very similar to the optimal reduction in the baseline calibration.

On the other hand, the denominator, i.e. the total assets in the banking sector, could be biased upwards as well if banks extend credit to each other, for example by holding bonds issued by other banks. To take this potential bias into account, we set $\phi^{SS} = 60\%$ and again recalibrate

$c_0 = 300$ to render ϕ^{SS} optimal. Naturally, the collateral premium is slightly larger in this case. The optimal collateral policy relaxation is comparable to the baseline: ϕ^{SS} declines to 52%.

7 Conclusion

This paper uses a quantitative DSGE model to evaluate how the net zero transition affects optimal central bank collateral policy. In this model, optimal collateral policy is just as lenient to balance the positive effects of increasing collateral supply to banks with central bank losses from exposure to risky collateral. Climate policy affects this trade-off in two ways. First, higher emission taxes increase the corporate default frequency in the short run if firms suddenly face an unsustainable leverage but not in the long run, as emission taxes do not make debt financing more or less attractive to firms. Second, emission taxes exert divestment and deleveraging pressure on non-financial firms by reducing the return on physical capital. Quantitatively, it is optimal to temporarily tighten collateral policy in response to emission tax shocks, while a structural relaxation of collateral policy improves welfare in the long run. These results are robust to various plausible modifications of the climate block, the collateral premium, and central bank losses from accepting risky debt as collateral.

References

- Annicchiarico, Barbara, Marco Carli, and Francesca Diluiso (2023). “Climate Policies, Macroprudential Regulation, and the Welfare Cost of Business Cycles.” CEIS Working Paper no. 543.
- Ashcraft, Adam, Nicolae Gârleanu, and Lasse Heje Pedersen (2011). “Two Monetary Tools: Interest Rates and Haircuts.” *NBER Macroeconomics Annual* 25(1), 143–180.
- Bekkum, Sjoerd van, Marc Gabarro, and Rustom M Irani (2018). “Does a Larger Menu Increase Appetite? Collateral Eligibility and Credit Supply.” *Review of Financial Studies* 31(3), 943–979.
- Berthold, Brendan, Ambrogio Cesa-Bianchi, Federico Di Pace, and Alex Haberis (2023). “The Heterogeneous Effects of Carbon Pricing: Micro and Macro Evidence.” CEPR Discussion Paper 18312.
- Bianchi, Javier, and Saki Bigio (2022). “Banks, Liquidity Management and Monetary Policy.” *Econometrica* 90(1), 391–454.
- Bilal, Adrien, and Diego Kaenzig (2025). “Does Unilateral Decarbonization Pay For Itself?” Working Paper.
- Bindseil, Ulrich, and Francesco Papadia (2006). “Credit Risk Mitigation in Central Bank Operations and its Effects on Financial Markets: The Case of the Eurosystem.” *ECB Occasional Paper* 49.
- Carattini, Stefano, Givi Melkadze, and Garth Heutel (2023). “Climate Policy, Financial Frictions, and Transition Risk.” *Review of Economic Dynamics* 51, 778–794.

- Cassola, Nuno, and François Koulischer (2019). “The Collateral Channel of Open Market Operations.” *Journal of Financial Stability* 41, 73–90.
- Chen, Hui, Zhuo Chen, Zhiguo He, Jinyu Liu, and Rengming Xie (2023). “Pledgeability and Asset Prices: Evidence from the Chinese Corporate Bond Markets.” *Journal of Finance* 78(5), 2563–2620.
- Cúrdia, Vasco, and Michael Woodford (2011). “The Central Bank Balance Sheet as an Instrument of Monetary Policy.” *Journal of Monetary Economics* 58(1), 54–79.
- De Fiore, Fiorella, Marie Hoerova, Ciaran Rogers, and Harald Uhlig (2024). “Money Markets, Collateral and Monetary Policy.” *BIS Working Paper* 997.
- Economides, George, and Anastasios Xepapadeas (2025). “Monetary Policy Stabilization in a New Keynesian Model Under Climate Change.” *Review of Economic Dynamics* 56, 101260.
- Environmental Defense Fund (2021). “Marginal Abatement Cost Curves for U.S. Net-Zero Energy Systems: A Systems Approach.” Technical Report.
- Fang, Hanming, Yongqin Wang, and Xian Wu (2025). “Collateral-Based Monetary Policy: Evidence from China.” *Working Paper*.
- Ferrari, Alessandro, and Valerio Nispi Landi (2023). “Toward a Green Economy: the Role of Central Bank’s Asset Purchases.” *International Journal of Central Banking* 19(5), 287–340.
- Fornaro, Luca, Veronica Guerreri, and Lucrezia Reichlin (2024). “Monetary Policy for the Green Transition.” Working Paper.
- Giovanardi, Francesco, and Matthias Kaldorf (2024). “Pro-Cyclical Emissions, Real Externalities, and Optimal Monetary Policy.” Bundesbank Discussion Paper.
- Giovanardi, Francesco, Matthias Kaldorf, Lucas Radke, and Florian Wicknig (2023). “The Preferential Treatment of Green Bonds.” *Review of Economic Dynamics* 51, 657–676.
- Gomes, João, Urban Jermann, and Lukas Schmid (2016). “Sticky Leverage.” *American Economic Review* 106(12), 3800–3828.
- Goncharov, Igor, Vasso Ioannidou, and Martin Schmalz (2023). “(Why) Do Central Banks Care about Their Profits?” *Journal of Finance* 78(5), 2991–3045.
- Hall, Robert E, and Ricardo Reis (2015). “Maintaining Central-Bank Financial Stability under New-Style Central Banking.” *NBER Working Paper* 21173.
- Heutel, Garth (2012). “How Should Environmental Policy Respond to Business Cycles? Optimal Policy under Persistent Productivity Shocks.” *Review of Economic Dynamics* 15(2), 244–264.
- Hüttl, Pia, and Matthias Kaldorf (2024). “Central Banks and Financial Integration: Evidence from the Eurosystem Collateral Framework.” *DIW Berlin Discussion Paper* (12).
- Jungherr, Joachim, and Immo Schott (2022). “Slow Debt, Deep Recessions.” *American Economic Journal: Macroeconomics* 14(1), 224–259.
- Koulischer, François, and Daan Struyven (2014). “Central bank liquidity provision and collateral quality.” *Journal of Banking and Finance* 49, 113–130.
- Kuehn, Lars-Alexander, and Lukas Schmid (2014). “Investment-Based Corporate Bond Pricing.” *Journal of Finance* 69(6), 2741–2776.

- McKibbin, Warwick J, Adele C Morris, Peter J Wilcoxon, and Augustus J Panton (2020). “Climate Change and Monetary Policy: Issues for Policy Design and Modelling.” *Oxford Review of Economic Policy* 36(3), 579–603.
- Mésonnier, Jean-Stéphane, Charles O’Donnell, and Olivier Toutain (2021). “The Interest of Being Eligible.” *Journal of Money, Credit and Banking* 54(2-3), 425–458.
- Nakov, Anton, and Carlos Thomas (2023). “Climate-Conscious Monetary Policy.” Working Paper.
- Pelizzon, Loriana, Max Riedel, Zorka Simon, and Marti G. Subrahmanyam (2024). “Collateral Eligibility of Corporate Debt in the Eurosystem.” *Journal of Financial Economics* 153, 103777.
- Schmitt-Grohé, Stephanie, and Martín Uribe (2007). “Optimal Simple and Implementable Monetary and Fiscal Rules.” *Journal of Monetary Economics* 54(6), 1702–1725.
- Tabakis, Evangelos, and Kentaro Tamura (2013). “The Use of Credit Claims as Collateral for Eurosystem Credit Operations.” *ECB Occasional Paper Series* (148).

A Additional Quantitative Results

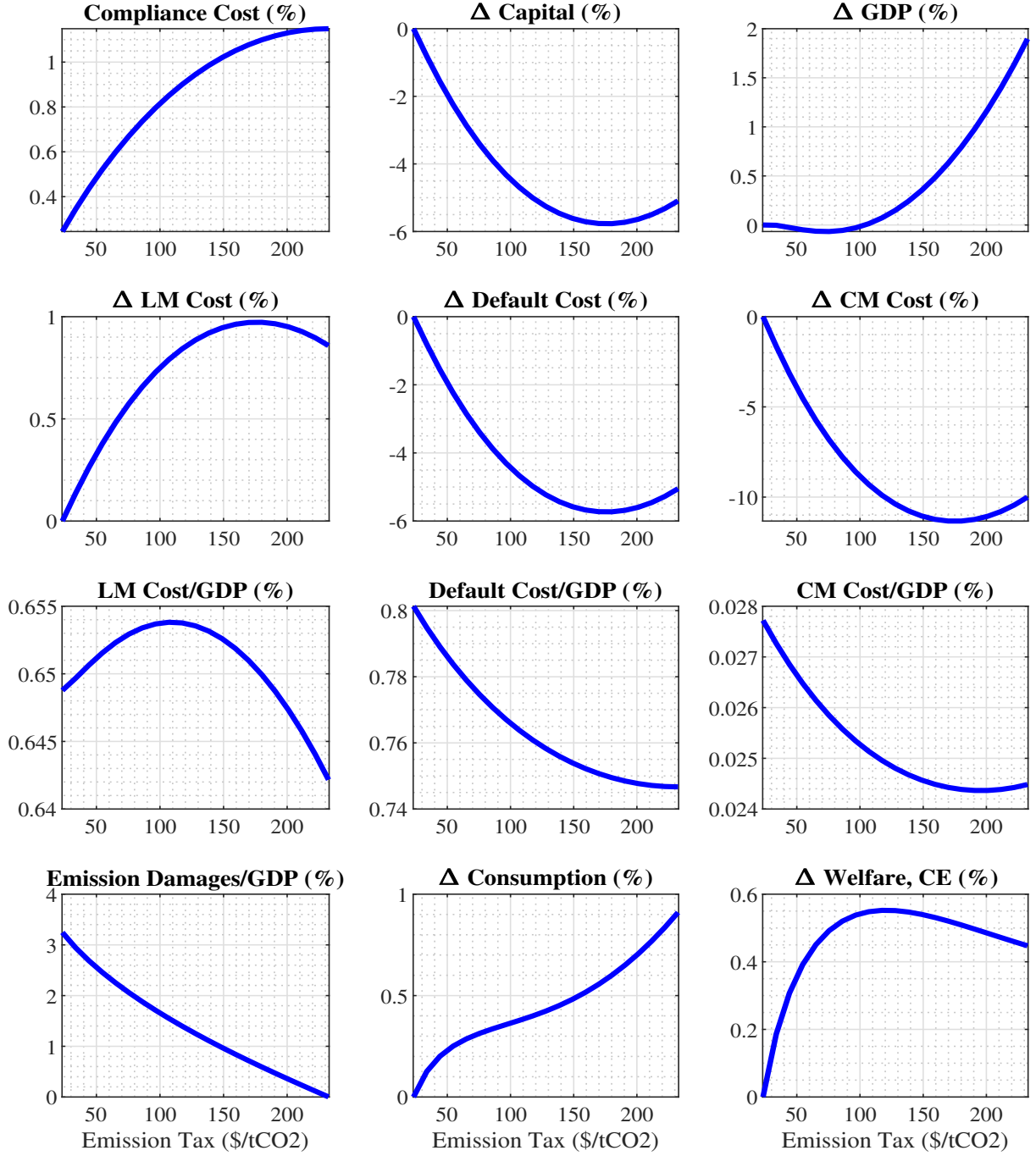


Figure A.1: Long Run Effects of Emission Taxes. This figure displays the effects of long run emission taxes on additional welfare relevant aggregates. First row: Climate policy compliance cost ξ_t per unit of capital, physical capital and GDP (both in % relative to the tax-free benchmark). Second row: liquidity management cost (5), resource losses of default (18) and collateral management cost (24) (all in % relative to the tax-free benchmark). Third row: cost functions relative to GDP, in percentage points. Fourth row: climate damages $1 - \exp\{\Psi_{Ee_t}\}$ in percentage points and consumption (in % relative to the tax-free benchmark). The welfare change in the bottom right is computed relative to the long run with the baseline collateral parameter of $\phi^{SS} = 65\%$ (bottom right). The emission tax on the x-axis is expressed in \$/tCO2.

Figure A.1 displays the effects of long run emission taxes of additional welfare relevant

macroeconomic aggregates. The upper row provides additional details on the transmission of higher emission taxes to the real economy. The per-unit cost of complying with climate policy (9), i.e. the sum of the carbon tax bill and the abatement spending, increase in the tax rate. Under full abatement, the tax bill is zero and compliance cost consist of the resources spent on abatement, which amount to more than one percent of GDP. The bottom left panel shows that emission damages decline by around 3% of GDP in this case. The gain from lower damages drastically increases abatement costs, such that effective TFP increases per the aggregate production function (26). Consequently, aggregate output is considerably larger under full abatement.

Quantitatively, net zero taxes reduce total capital by around 5%. The effect is much more pronounced in the robustness check without climate damages, as only the negative effect of abatement costs on the investment return is present. In the baseline model the positive effect of emission taxes on climate damages, consumption increases by around one percent in the long run. The welfare gain is substantial, but slightly diminishes for emission taxes close to net zero: going the "last mile" to net zero is very costly due to the convex abatement cost. However, this has positive but comparatively small effects on climate damages, which are convex in emissions.

The second row of Figure A.1 show how banks' liquidity management cost, resource losses from corporate default, and collateral management cost respond to climate policy. The shape of these cost functions in the second row essential traces the effect of emission taxes on the change in aggregate capital. Any reduction in capital goes hand in hand with a reduction in debt issuance, which increases banks' liquidity management cost $\mathcal{L}(\bar{b}_t)$. It mechanically reduces the cost of corporate default $\mathcal{F}_t$ and the collateral management cost $\mathcal{C}_t$ incurred by the central bank. The third row shows the different cost functions relative to GDP. Due to the secular increase in aggregate GDP, the relative importance of all three cost components declines as emission taxes achieve the net zero level.

In Figure A.2, we show how the same macroeconomic aggregates respond to changes in the collateral parameter ϕ^{SS} . We consider the case with the initial, low emission tax of 23\$/tCO₂ (dashed red line) and with full abatement taxes (230\$/tCO₂, dotted green line). Importantly, the transmission of collateral policy on macroeconomic aggregates is largely independent of the climate policy stance. For example, aggregate investment would exhibit the same response to a reduction of ϕ^{SS} by 10 percentage points, irrespective of the emission tax.

Naturally, the per unit cost of complying with climate policy do not depend on the collateral policy regime in place. Consequently, climate damages are not affected either. However, the stance of collateral policy has a macroeconomically relevant effect on aggregate investment and output. Relative to the baseline with $\phi^{SS} = 65\%$, capital would increase by more than 0.5% if all assets would be accepted without valuation haircuts. GDP would expand by around 0.25%. These investment effects stem from the higher prices of debt securities if eligibility is high/haircuts are low.

As the second row shows, this reduces the liquidity management cost for banks but at the same time increases resource losses from default in the non-financial sector and strongly increases the collateral management cost incurred by the central bank. Consumption largely follows the pattern of aggregate GDP, although it starts declining for very lenient policy, i.e. when default

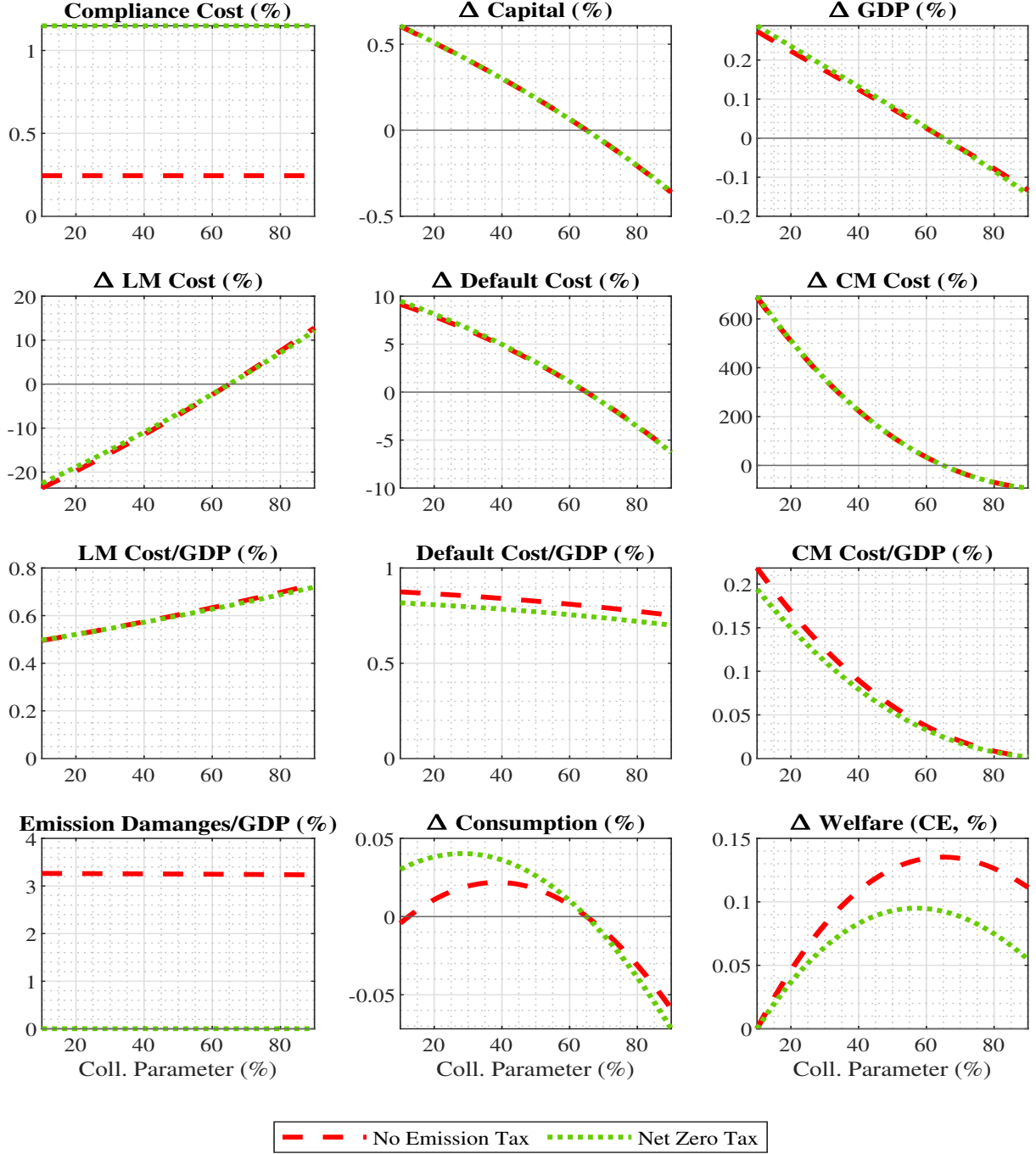


Figure A.2: Long Run Effects of Collateral Policy. This figure displays the effects of long run collateral policy on additional welfare relevant aggregates. First row: Climate policy compliance cost ξ_t per unit of capital, physical capital and GDP (both in % relative to the tax-free benchmark). Second row: liquidity management cost (5), resource losses of default (18) and collateral management cost (24) (all in % relative to the tax-free benchmark). Third row: cost functions relative to GDP, in percentage points. Fourth row: climate damages $1 - \exp\{\Psi_{Ee_t}\}$ in percentage points and consumption (in % relative to the tax-free benchmark). The welfare change in the bottom right is computed relative to the long run with the baseline collateral parameter of $\phi^{SS} = 65\%$ (bottom right). The emission tax on the x-axis is expressed in $\$/tCO_2$. The dashed red line corresponds to low emission taxes of $23\$/tCO_2$ while the dotted green line refers to emission taxes consistent with net zero.

costs become large. The bottom right panel shows how long run collateral policy affects welfare. The maximum is attained for a lower collateral parameter in the full abatement case than

in the long run equilibrium with low emission taxes. Again, this is not due to a change in the *transmission* of collateral policy to macro aggregates due to emission taxes, but because emission taxes have permanent effects on macro aggregates that are relevant for the optimal collateral framework.

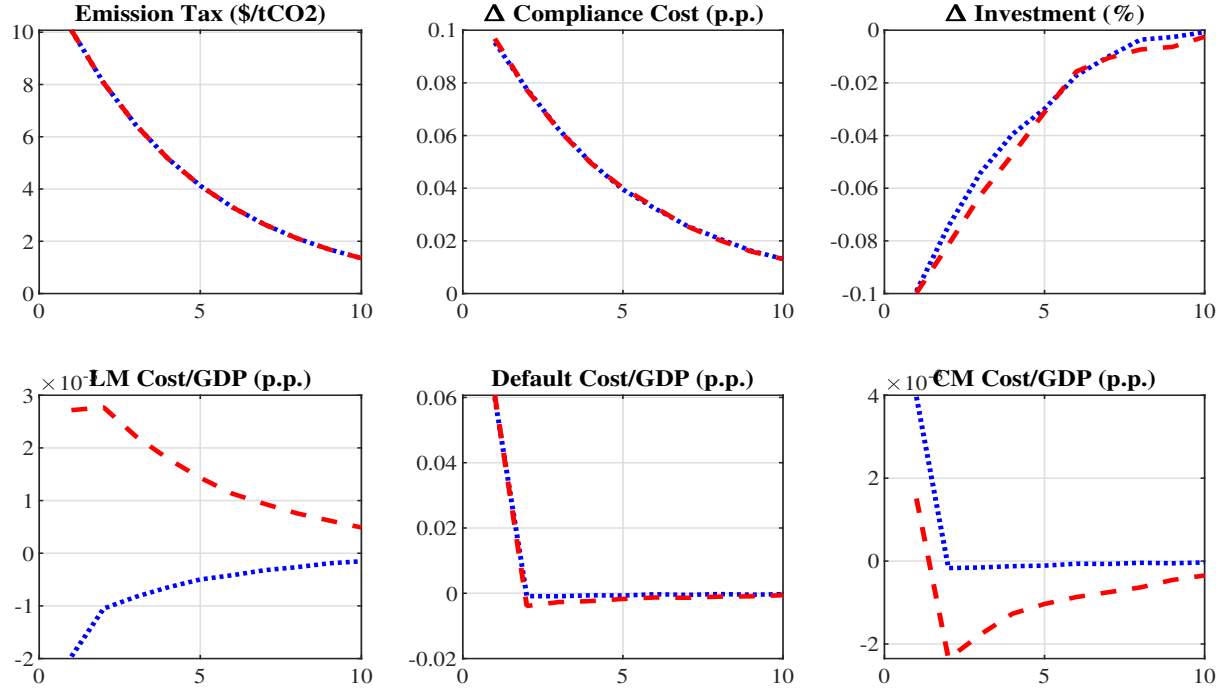


Figure A.3: Short Run Effects of Emission Taxes. This figure displays the effects of long run emission taxes on additional welfare relevant aggregates. First row: Climate policy compliance cost ξ_t per unit of capital and physical capital and GDP (both in % relative to the tax-free benchmark). Second row: liquidity management cost (5), resource losses of default (18) and collateral management cost (24) (all in % relative to the tax-free benchmark). Time on the x-axis is expressed in years.